ECON 6018 TUTORIAL

- By: Teresa Ng
- Lesson Tutorial
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Footnote: The practice questions are adapted from Essential Mathematics for Economic Analysis (5th Edition) by Knut Sydsaeter, Peter Hammond, Arne Strom, and Andres Carvajal. All credit for the original questions belongs to the authors. The solutions and explanations provided here are my own work.

1. Inverse Matrix

From Lect:

Important things to note:

a. All Inverse Matrix are square matrix

b. The determinants must be > 0

Properties of Inverse

a. $AA^{-1} = A^{-1}A = I$

b. $Ax = d$

$x = A^{-1}(d)$

Qns:

(1) Given that A, B are $n \times n$ matrices such that $AB = 0$.

Prove that if A is invertible then B is not invertible

(2) Prove that $(AB)^{-1} = B^{-1}A^{-1}$

(3) Prove that if $BA = I$ then $BA = AB$

(2) $(AB)^{-1} = B^{-1}A^{-1}$

$(AB)(AB)^{-1} = I$

$AB\,B^{-1}A^{-1} = AIA^{-1}$

$= I$

Therefore : $(AB)^{-1} = B^{-1}A^{-1}$

(3) $BA = I$

$BAA^{-1} = IA^{-1}$

$B = A^{-1} \longrightarrow AB = AA^{-1} = I = BA$

$B^{-1}BA = B^{-1}I$ or

$A = B^{-1} \longrightarrow AB = B^{-1}B = I = BA$

2. Determinant

Qns: Determinant of order 2 (16.1)

(1) Find the determinants

a. $\begin{vmatrix} 3 & 0 \\ 2 & 6 \end{vmatrix}$ b. $\begin{vmatrix} a & a \\ b & b \end{vmatrix}$ c. $\begin{vmatrix} a+b & a-b \\ a-b & a+b \end{vmatrix}$ (d) $\begin{vmatrix} 3^t & 2^t \\ 3^{t-1} & 2^{t-1} \end{vmatrix}$

(2) Find a and b such that the $tr(A) = 0$ and $|A| = -10$ whereby matrix $A = \begin{pmatrix} a & 3 \\ b & 1 \end{pmatrix}$

Answer:

1(a) $\begin{vmatrix} 3 & 0 \\ 2 & 6 \end{vmatrix} = 18 - 0 = 18$

1(b) $\begin{vmatrix} a & a \\ b & b \end{vmatrix} = 0$

1(c) $\begin{vmatrix} (a+b) & (a-b) \\ (a-b) & (a+b) \end{vmatrix} = 4ab$

1(d) $\begin{vmatrix} 3^t & 2^t \\ 3^{t-1} & 2^{t-1} \end{vmatrix} = 3^t 2^t 2^{-1} - 3^t 2^t 3^{-1}$

$= (3^t 2^t)(2^{-1} - 3^{-1})$

$= \frac{1}{6}(3^t 2^t) = \frac{6^t}{6} = 6^{t-1}$

(2) $a+1=0 \quad a=-1$

$\begin{vmatrix} -1 & 3 \\ b & 1 \end{vmatrix} = -1-3b = -10$

$-3b = -9$

$b = 3$

Qns: Determinants of Order 3 (16.2)

(1)

a. $\begin{vmatrix} 1 & -1 & 0 \\ 1 & 3 & 2 \\ 1 & 0 & 0 \end{vmatrix}$ b. $\begin{vmatrix} 1 & -1 & 0 \\ 1 & 3 & 2 \\ 1 & 2 & 1 \end{vmatrix}$ c. $\begin{vmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{vmatrix}$ d. $\begin{vmatrix} a & 0 & b \\ 0 & e & 0 \\ c & 0 & d \end{vmatrix}$

(2) Given,

$$A_t = \begin{pmatrix} 1 & t & 0 \\ -2 & -2 & -1 \\ 0 & 1 & t \end{pmatrix}$$

Find the determinant

Ans

1(a) $\begin{vmatrix} 1 & -1 & 0 \\ 1 & 3 & 2 \\ 1 & 0 & 0 \end{vmatrix} = -1(-1)^3 \begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix}$

$= (-2)$

$= -2$

1(b) $\begin{vmatrix} 1 & -1 & 0 \\ 1 & 3 & 2 \\ 1 & 2 & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix}$

$= -2$

$1(-1)^2 \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} + (-1)(-1)^3 \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix}$

$l(c)$ adf

$l(d) = aed$

So how does this apply to inverse ☆

$$A^{-1} = \frac{1}{|A|} \begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{pmatrix}^T$$

↓

Adj(A)

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$ for 2x2

$|A| = 0$ no solution or infinite solⁿ

$|A| > 0$, unique solution

(1) Find the solⁿ to

$$\begin{aligned} 3x_1 + x_2 &= 1 \\ x_1 - x_2 + 2x_3 &= 2 \\ 2x_1 + 3x_2 - x_3 &= 3 \end{aligned}$$

Sol$^{\underline{n}}$

Step 1: Put them in terms of $Ax = d$

$$\begin{pmatrix} 3 & 1 & 0 \\ 1 & -1 & 2 \\ 2 & 3 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

Step 2

$$|A| = \begin{vmatrix} 3 & 1 & 0 \\ 1 & -1 & 2 \\ 2 & 3 & -1 \end{vmatrix} = -15 + 5 = -10$$

Step 3

$$|B| = \begin{vmatrix} 1 & 1 & 0 \\ 2 & -1 & 2 \\ 3 & 3 & -1 \end{vmatrix} = -3$$

$$|C| = \begin{vmatrix} 3 & 1 & 0 \\ 1 & 2 & 2 \\ 2 & 3 & -1 \end{vmatrix} = 19$$

$$|D| = \begin{vmatrix} 3 & 1 & 1 \\ 2 & -1 & 2 \\ 1 & 3 & 3 \end{vmatrix} = -21$$

$$x_1 = \frac{|B|}{|A|} = 0.3 \qquad x_2 = \frac{|C|}{|A|} = 1.9 \qquad x_3 = \frac{|D|}{|A|} = 2.1$$

(2) $Y - C = A_0$ (Chpt 16.2)

$-bY + C + bT = a$

$-tY + T = d$

Sdn :

$$\begin{pmatrix} 1 & -1 & 0 \\ -b & 1 & b \\ -t & 0 & 1 \end{pmatrix} \begin{pmatrix} Y \\ C \\ T \end{pmatrix} = \begin{pmatrix} A_0 \\ a \\ d \end{pmatrix}$$

$|A| = 1 - b - tb$

$|B| = A_0 + a - db$

$|C| = a - db + A_0 b - A_0 tb$

$|D| = at - bd + d + A_0 t$

(3) Given

$$x + y - z = 1$$
$$x - y + 2z = 2$$
$$x + 2 + az = b$$

For what value of a is there a unique sol^n

Ans:

$$\begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 2 \\ 1 & 2 & a \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ b \end{pmatrix}$$

$$\begin{vmatrix} 1 & 1 & -1 \\ 1 & -1 & 2 \\ 1 & 2 & a \end{vmatrix} = -a-4-(a-2)+(-1)(2+1)$$

$$= -2a-4+2-3$$

$$= -2a-5$$

For unique

$$-2a-5 \neq 0$$

$$-2a \neq 5$$

$$a \neq -\frac{5}{2}$$

(4) Given

$$\begin{pmatrix} a & 1 & (a+1) \\ 1 & 2 & 1 \\ 3 & 4 & 7 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

For what value of a is there a unique sol^n